1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 analysis highlights a tension: MSH6 deficiency should raise neoantigen burden and attract anti-tumor immunity, yet many tumors remain clinically aggressive — why does high antigenicity not always translate into better outcomes?

Working Hypothesis

Does high stromal-to-tumor ratio (TSR), driven by TGFB1 signaling, immunosuppress the anti-tumor immune response in bladder tumors that harbor MSH6 deficiency, thereby negating the expected OS advantage conferred by increased neoantigen burden? If true, then among MSH6-deficient bladder tumors, those with high TSR will have worse OS than those with low TSR; antifibrotic or TGF-β–targeted interventions could restore immune infiltration and improve OS in the high-TSR subset. This hypothesis posits a mechanistic chain: (i) MSH6 deficiency elevates neoantigen load; (ii) TGFB1-driven stromal remodeling increases TSR and fosters an immune-excluding microenvironment; (iii) the net effect is reduced effective anti-tumor immunity and poorer survival in high-TSR, MSH6-deficient tumors; (iv) disruption of TGFB1 signaling or stromal remodeling reverses immune exclusion and improves survival. A secondary modifier, GREM1, may reflect stromal remodeling status and further modulate this axis.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TGFB1, MSH6
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGFB1: TGFB1 (TGF-β signaling → fibrotic, immune-excluding stroma) and MSH6 (MMR loss → increased neoantigen load).

MSH6: MSH6 deficiency should raise neoantigen burden and attract anti-tumor immunity, yet many tumors remain clinically aggressive — why does high antigenicity not always translate into better outcomes? (2) Tier 1 paths used: I built on TGFB1 (TGF-β signaling → fibrotic, immune-excluding stroma) and MSH6 (MMR loss → increased neoantigen load).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CTCs	Supporting modifier from knowledge graph

Secondary	GREM1	Supporting modifier from knowledge graph
Secondary	Immunoscore	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CTCs: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- GREM1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.